

# Series with non-negative terms



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# Comparison test



# Introduction: Comparison test

- ▶ In this chapter we study **series with non-negative terms**, for which we have very effective tools to determine their **behavior**
- ▶ The first tool is the **comparison test**: we compare the series under study with a series whose behavior we already know
- ▶ As a first application, we will prove that the series  $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$  diverges for  $\alpha \leq 1$

## Series with non-negative terms

- ▶ A series  $\sum a_k$  **with non-negative terms** (or positive terms) is **regular**, that is, it is either convergent or divergent to  $+\infty$  (it cannot be irregular, as we have proved)
- ▶ Such a series converges if and only if the sequence of its partial sums is bounded

# Comparison test

## Theorem 1: Comparison test

*Let  $\{a_k\}$  and  $\{b_k\}$  be two sequences with non-negative terms such that  $a_k \leq b_k$  eventually; then:*

$$i) \quad \sum b_k \text{ convergent} \Rightarrow \sum a_k \text{ convergent}$$

$$ii) \quad \sum a_k \text{ divergent} \Rightarrow \sum b_k \text{ divergent}$$

- ▶ The series  $\sum b_k$  is called the **dominating series** (majorant), while the series  $\sum a_k$  is called the **dominated series** (minorant).

## Proof (1/2)

Since  $a_k \leq b_k$  eventually, there exists  $m \in \mathbb{N}$  such that:

$$a_k \leq b_k, \quad \forall k \geq m$$

Let us now consider the  $n$ -th partial sums of the tails of the two sequences  $\{a_k\}$  and  $\{b_k\}$ , with  $n > m$ :

$$s_n^a = \sum_{k=m+1}^n a_k \quad \text{and} \quad s_n^b = \sum_{k=m+1}^n b_k$$

Since  $0 \leq a_k \leq b_k$ ,  $\forall k \geq m$ , we have

$$0 \leq s_n^a \leq s_n^b, \quad \forall n \geq m \quad (*)$$

## Proof (2/2)

- ▶ The sequences  $\{s_n^a\}$  and  $\{s_n^b\}$  are regular, since  $\{a_k\}$  and  $\{b_k\}$  have non-negative terms. Hence the claims *i*) and *ii*) are logically equivalent, so it suffices to prove the second one.
- ▶ Saying that  $\sum a_n$  is divergent means, by the definition of divergent series, that  $s_n \rightarrow +\infty$  as  $n \rightarrow +\infty$ . Moreover, since every tail has the same behavior as the original series, we have  $s_n^a \rightarrow +\infty$  as  $n \rightarrow +\infty$ .
- ▶ From (\*), by the comparison theorem for sequences, also  $s_n^b \rightarrow +\infty$  as  $n \rightarrow +\infty$ . Hence the tail of  $\{b_k\}$  is divergent and consequently  $\sum b_n$  is divergent.



The series  $\sum 1/k^\alpha$  with  $\alpha \leq 1$

### Observation 1

*Given  $\alpha \leq 1$ , the series  $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$  is divergent to  $+\infty$ .*

# Proof

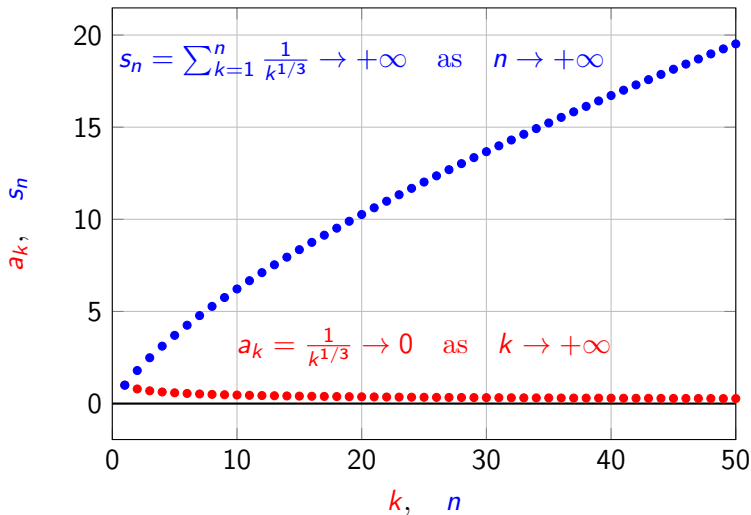
With  $\alpha = 1$  we have the harmonic series, which diverges to  $+\infty$ .

With  $\alpha < 1$ , the series dominates the harmonic series, since:

$$\frac{1}{k} \leq \frac{1}{k^\alpha}, \quad \forall k \in \mathbb{N}, k \geq 1 \quad \text{and} \quad \alpha < 1$$

hence  $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$  is divergent by the comparison test. □

Graphical interpretation:  $\alpha = \frac{1}{3}$



# Summary

## Series with non-negative terms:

- ▶ They are always regular: they converge or diverge to  $+\infty$
- ▶ They converge if and only if the sequence of partial sums is bounded

## Comparison test:

- ▶ If  $0 \leq a_k \leq b_k$  eventually: convergent dominating series  $\Rightarrow$  convergent dominated series; divergent dominated series  $\Rightarrow$  divergent dominating series
- ▶ Consequence:  $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$  diverges to  $+\infty$  for  $\alpha \leq 1$

# Limit comparison test



# Introduction: Limit comparison test

- ▶ It is often not easy to find an inequality  $a_k \leq b_k$ , while it is simple to find a sequence **asymptotically equivalent** to the general term
- ▶ The **limit comparison test** states that two series with positive terms and asymptotically equivalent general terms have the **same behavior**
- ▶ We will use it to prove that  $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$  converges for  $\alpha \geq 2$  and to study a first example

## Theorem 2: Limit comparison test

*Let  $\{a_k\}$  and  $\{b_k\}$  be two sequences with positive terms. If the sequences are asymptotically equivalent, that is, if:*

$$a_k \sim b_k \quad \text{as } k \rightarrow +\infty$$

*then the corresponding series  $\sum a_k$  and  $\sum b_k$  are regular and have the same behavior, i.e., either they are both convergent or they are both divergent.*

## Proof (1/2)

The series  $\sum a_k$  and  $\sum b_k$  are regular because the sequences  $\{a_k\}$  and  $\{b_k\}$  have positive terms. Since  $a_k \sim b_k$  as  $k \rightarrow +\infty$ , we have:

$$\frac{a_k}{b_k} \rightarrow 1 \quad \text{as } k \rightarrow +\infty$$

Hence, for every  $\varepsilon > 0$ , there exists  $m \in \mathbb{N}$  such that  $\forall k \geq m$  we have:

$$1 - \varepsilon < \frac{a_k}{b_k} < 1 + \varepsilon \quad \text{and hence} \quad (1 - \varepsilon) b_k < a_k < (1 + \varepsilon) b_k$$

since  $b_k > 0$ ,  $\forall k$ .

## Proof (2/2)

- ▶ We have thus proved that  $(1 - \varepsilon) b_k < a_k < (1 + \varepsilon) b_k$  eventually.
- ▶ Hence, by the comparison test, the series  $\sum a_k$  and  $\sum b_k$  have the same behavior.
- ▶ The first of the two inequalities implies that if  $\sum a_k$  is convergent then  $\sum b_k$  is also convergent, while the second one implies that if  $\sum a_k$  is divergent then  $\sum b_k$  is also divergent.



The series  $\sum 1/k^\alpha$  with  $\alpha \geq 2$

### Observation 2

*Given  $\alpha \geq 2$ , the series  $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$  is convergent.*

# Proof

With  $\alpha = 2$  we have:

$$\frac{1}{k^2} \sim \frac{1}{k(k+1)} \quad \text{as } k \rightarrow +\infty$$

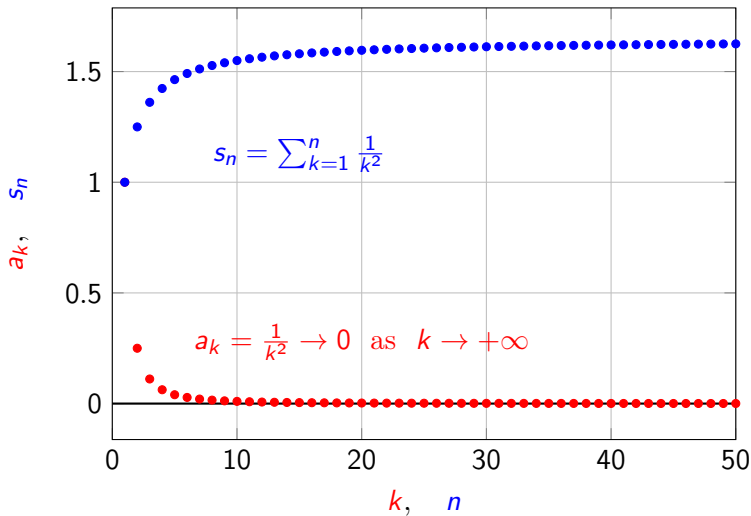
and the series  $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$  converges (Mengoli's series). Hence, by the limit comparison test,  $\sum_{k=1}^{\infty} \frac{1}{k^2}$  also converges.

With  $\alpha > 2$ , the series is dominated by the series  $\sum_{k=1}^{\infty} \frac{1}{k^2}$ , since:

$$\frac{1}{k^\alpha} \leq \frac{1}{k^2}, \quad \forall k \in \mathbb{N}, k \geq 1 \quad \text{and} \quad \alpha > 2$$

hence  $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$  is convergent by the comparison test. □

## Graphical interpretation: $\alpha = 2$



## Example (1/2)

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{5k + \cos k}{3 + 2k^3}$$

We have:

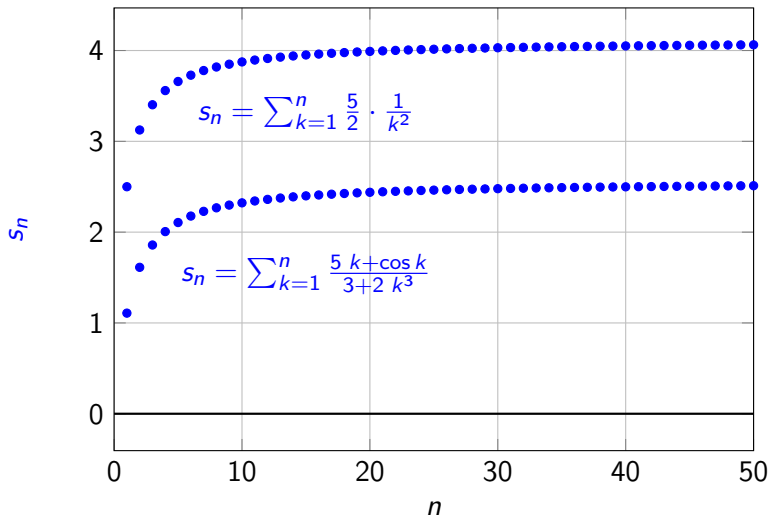
$$\frac{5k + \cos k}{3 + 2k^3} \sim \frac{5}{2} \cdot \frac{1}{k^2} \quad \text{as } k \rightarrow +\infty$$

and

$$\sum_{k=1}^{\infty} \frac{5}{2} \cdot \frac{1}{k^2} \quad \text{converges}$$

hence the series is convergent by the limit comparison test.

## Example (2/2)



# Warning

However, even if the sequences are asymptotically equivalent, the corresponding series may not have the same sum.

- ▶ In the previous example, both series converge but their sums are different.

# Summary

## Limit comparison test:

- ▶ If  $a_k, b_k > 0$  and  $a_k \sim b_k$  as  $k \rightarrow +\infty$ , then  $\sum a_k$  and  $\sum b_k$  have the same behavior
- ▶ The sums of the two series may however be different

## Consequences:

- ▶  $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$  converges for  $\alpha \geq 2$  (comparison with Mengoli's series)
- ▶ To study a series it suffices to find a simpler asymptotically equivalent term

# Condensation test

# Introduction: Condensation test

- ▶ So far we know that  $\sum \frac{1}{k^\alpha}$  diverges for  $\alpha \leq 1$  and converges for  $\alpha \geq 2$ : the case  $1 < \alpha < 2$  remains open
- ▶ The **condensation test** replaces the series  $\sum a_k$  with the “condensed” series  $\sum 2^k a_{2^k}$ , which only uses the terms with index  $2^k$
- ▶ For decreasing terms the two series have the same behavior: the proof relies on two inequalities between partial sums, proved by induction

## Theorem 3: Condensation test

*If  $\{a_k\}$  is an eventually decreasing sequence with non-negative terms, then the series*

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \sum_{k=0}^{\infty} 2^k a_{2^k}$$

*are regular and have the same behavior, i.e., either they are both convergent or they are both divergent.*

## Partial sums $\tilde{s}_n$ and $s_n$

- ▶ Before proving the theorem, let us consider the sequence  $\{\tilde{s}_n\}$  of the partial sums of the sequence  $\{2^k a_{2^k}\}$ , that is, the sequence:

$$\tilde{s}_n = \sum_{k=0}^n 2^k a_{2^k} \quad \forall n \geq 0$$

- ▶ Let us derive two important relations with the sequence  $\{s_n\}$  of the partial sums of the sequence  $\{a_k\}$ , that is, the sequence:

$$s_n = \sum_{k=1}^n a_k \quad \forall n \geq 1$$

## Partial sums $s_{2^n-1}$

- We observe that, for certain values of  $n$ , the partial sums  $s_n$  are:

$$\underbrace{s_1}_{= s_{2^1-1}} = a_1$$

$$\underbrace{s_3}_{= s_{2^2-1}} = s_1 + a_2 + \underbrace{a_3}_{\leq a_2} \leq a_1 + 2 a_2$$

$$\underbrace{s_7}_{= s_{2^3-1}} = s_3 + a_4 + \underbrace{a_5}_{\leq a_4} + \underbrace{a_6}_{\leq a_4} + \underbrace{a_7}_{\leq a_4} \leq a_1 + 2 a_2 + 4 a_4$$

$$\underbrace{s_{15}}_{= s_{2^4-1}} = s_7 + a_8 + \underbrace{a_9}_{\leq a_8} + \cdots + \underbrace{a_{15}}_{\leq a_8} \leq a_1 + 2 a_2 + 4 a_4 + 8 a_8$$

# First inequality

## Observation 3

*Given an eventually decreasing sequence  $\{a_k\}$  with non-negative terms, we have:*

$$s_{2^n-1} \leq \underbrace{\sum_{k=0}^{n-1} 2^k a_{2^k}}_{=\tilde{s}_{n-1}} \quad \forall n \geq 1 \quad (\text{P1})$$

## Proof (1/2)

We prove by induction on  $2^n$  that

$$s_{2^n-1} \leq \sum_{k=0}^{n-1} 2^k a_{2^k} \quad \forall n \geq 1$$

**Base case.** Let  $n = 1$ . Then the claim becomes

$$s_{2^1-1} \leq 2^0 a_1$$

i.e.,  $a_1 \leq a_1$ , which is clearly true.

## Proof (2/2)

**Inductive step.** Assume that it is true for  $2^{n-1}$  and let us prove it for  $2^n$ . By the inductive hypothesis, we have

$$s_{2^{n-1}-1} \leq \sum_{k=0}^{n-2} 2^k a_{2^k}$$

Moreover, we have:

$$\begin{aligned} s_{2^n-1} &= s_{2^{n-1}-1} + \underbrace{\sum_{i=2^{n-1}}^{2^n-1} a_i}_{\leq 2^{n-1} a_{2^{n-1}}} \\ &\leq 2^{n-1} a_{2^{n-1}} \end{aligned}$$

hence

$$s_{2^n-1} \leq \sum_{k=0}^{n-2} 2^k a_{2^k} + 2^{n-1} a_{2^{n-1}} = \sum_{k=0}^{n-1} 2^k a_{2^k}$$

which is exactly the desired claim for  $2^n$ .



## Partial sums $\tilde{s}_n$

- We now observe that the values of  $\tilde{s}_n$  are:

$$\tilde{s}_0 = a_1 \leq 2 a_1 = 2 \underbrace{s_1}_{= s_{2^0}}$$

$$\tilde{s}_1 = \tilde{s}_0 + 2 a_2 \leq 2 s_1 + 2 a_2 = 2 \underbrace{s_2}_{= s_{2^1}}$$

$$\tilde{s}_2 = \tilde{s}_1 + 4 a_4 \leq 2 s_2 + 2 a_3 + 2 a_4 = 2 \underbrace{s_4}_{= s_{2^2}}$$

(since  $a_4 \leq a_3$ )

$$\tilde{s}_3 = \tilde{s}_2 + 8 a_8 \leq 2 s_4 + 2 a_5 + \cdots + 2 a_8 = 2 \underbrace{s_8}_{= s_{2^3}}$$

(since  $a_8 \leq a_7 \leq a_6 \leq a_5$ )

## Second inequality

### Observation 4

*Given an eventually decreasing sequence  $\{a_k\}$  with non-negative terms, we have:*

$$\underbrace{\sum_{k=0}^n 2^k a_{2^k}}_{=\tilde{s}_n} \leq 2 s_{2^n} \quad \forall n \geq 0 \quad (\text{P2})$$

## Proof (1/2)

We prove by induction on  $2^n$  that

$$\tilde{s}_n \leq 2 s_{2^n} \quad \forall n \geq 0$$

**Base case.** Let  $n = 0$ . Then the claim becomes

$$\tilde{s}_0 \leq 2 s_{2^0} = 2 s_1$$

i.e.,  $a_1 \leq 2 a_1$ , which is clearly true.

## Proof (2/2)

**Inductive step.** Assume that it is true for  $n - 1$  and let us prove it for  $n$ . By the inductive hypothesis, we have

$$\tilde{s}_{n-1} \leq 2 s_{2^{n-1}}$$

Moreover, we have:

$$\begin{aligned}\tilde{s}_n &= \tilde{s}_{n-1} + \underbrace{2^n a_{2^n}}_{2^n} \\ &\leq \sum_{i=2^{n-1}+1}^{2^n} 2 a_i\end{aligned}$$

hence

$$\tilde{s}_n \leq 2 s_{2^{n-1}} + \sum_{i=2^{n-1}+1}^{2^n} 2 a_i = 2 s_{2^n}$$

which is exactly the desired claim for  $n$ . □

We are now able to prove the condensation test.

## Proof (1/2)

From relation (P1) we have:

$$s_{2^n-1} \leq \tilde{s}_{n-1}, \quad \forall n \geq 1$$

- ▶ Hence, if  $\{\tilde{s}_n\}$  is convergent, that is, if  $\sum_{k=0}^{\infty} 2^k a_{2^k}$  is convergent, the subsequence  $\{s_{2^n-1}\}$  is bounded.
- ▶ Since  $\{s_n\}$  is monotone, the whole sequence  $\{s_n\}$  is bounded and convergent by the monotone sequence theorem.
- ▶ Consequently, if  $\sum_{k=0}^{\infty} 2^k a_{2^k}$  is convergent, then  $\sum_{k=1}^{\infty} a_k$  is also convergent.

## Proof (2/2)

From relation (P2) we have:

$$\tilde{s}_n \leq 2 s_{2^n}, \quad \forall n \geq 0$$

- ▶ Hence, if  $\{s_n\}$  is convergent, that is, if  $\sum_{k=1}^{\infty} a_k$  is convergent, then the subsequence  $\{s_{2^n}\}$  is convergent (since  $\{s_n\}$  is monotone). Consequently, by comparison,  $\{\tilde{s}_n\}$  is also convergent, that is,  $\sum_{k=0}^{\infty} 2^k a_{2^k}$  converges.
- ▶ Moreover, since  $\{a_k\}$  has non-negative terms, the series cannot be irregular; consequently, we also have:

$$\sum_{k=1}^{\infty} a_k \text{ divergent} \iff \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ divergent}$$

which completes the proof of the theorem.



# Necessary and sufficient condition

Given an eventually decreasing sequence  $\{a_k\}$  with non-negative terms, we have proved that:

$$\sum_{k=1}^{\infty} a_k \text{ conv./div.} \iff \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ conv./div.}$$

- Hence the convergence/divergence of  $\sum_{k=0}^{\infty} 2^k a_{2^k}$  is a necessary and sufficient condition for the convergence/divergence of  $\sum_{k=1}^{\infty} a_k$ .

# Summary

## Condensation test:

- ▶ If  $\{a_k\}$  is eventually decreasing with non-negative terms,  $\sum_{k=1}^{\infty} a_k$  and  $\sum_{k=0}^{\infty} 2^k a_{2^k}$  have the same behavior

## Idea of the proof:

- ▶ (P1):  $s_{2^n-1} \leq \tilde{s}_{n-1}$  and (P2):  $\tilde{s}_n \leq 2 s_{2^n}$ , proved by induction
- ▶ The partial sums are monotone: we conclude with the monotone sequence theorem and by comparison

# Generalized harmonic series



# Introduction: Generalized harmonic series

- ▶ The series  $\sum \frac{1}{k^\alpha}$  are the reference series most frequently used in comparisons
- ▶ Thanks to the condensation test we can now determine their behavior for **every**  $\alpha \in \mathbb{R}$
- ▶ The result reduces to the study of a **geometric series**

# Generalized harmonic series

## Definition 1: Generalized harmonic series

Given  $\alpha \in \mathbb{R}$ , the series

$$\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$$

is called the *generalized harmonic series*

# Behavior of the generalized harmonic series

## Theorem 4: Behavior of the generalized harmonic series

Given  $\alpha \in \mathbb{R}$ , the generalized harmonic series

$$\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}} \quad \text{is} \quad \begin{cases} \text{divergent to } +\infty & \text{if } \alpha \leq 1 \\ \text{convergent} & \text{if } \alpha > 1 \end{cases}$$

# Proof

We have already established that it diverges for  $\alpha \leq 1$ .

For  $\alpha > 1$ , the sequence  $\left\{ \frac{1}{k^\alpha} \right\}$  is a decreasing sequence with positive terms. We have:

$$\sum_{k=0}^{\infty} 2^k a_{2^k} = \sum_{k=0}^{\infty} 2^k \frac{1}{(2^k)^\alpha} = \sum_{k=0}^{\infty} (2^{1-\alpha})^k$$

that is, a geometric series with ratio  $2^{1-\alpha}$ , which converges if and only if  $2^{1-\alpha} < 1$ , that is, if  $1 - \alpha < 0$ .

Hence, for  $\alpha > 1$ , by the condensation test, the series converges.



# Summary

**Generalized harmonic series**  $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$ :

- ▶ It diverges to  $+\infty$  if  $\alpha \leq 1$  (comparison test with the harmonic series)
- ▶ It converges if  $\alpha > 1$  (condensation test: we obtain the geometric series with ratio  $2^{1-\alpha}$ )

**Use:**

- ▶ It is the reference series for the comparison and limit comparison tests

# Passing from the discrete to the continuous



# Introduction: Passing from the discrete to the continuous

- ▶ The tools used to establish asymptotic estimates of functions can also be used to obtain asymptotic estimates of sequences (**passing from the discrete to the continuous**)
- ▶ They provide useful tools for studying the behavior of a series with positive terms
- ▶ We see three examples in which asymptotic equivalences and Maclaurin expansions, combined with the limit comparison test, determine the behavior of the series

## Example

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{e^{1/k} - 1}{k}$$

From the asymptotic equivalence

$$e^{\varepsilon(x)} - 1 \sim \varepsilon(x) \quad \text{as} \quad \varepsilon(x) \rightarrow 0$$

we have

$$\frac{e^{1/k} - 1}{k} \sim \frac{1}{k^2} \quad \text{as} \quad k \rightarrow +\infty$$

Therefore the series, which has positive terms, converges by limit comparison with the series of  $1/k^2$ .

## Example

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \log \left( \frac{k+3}{k+2} \right)$$

We have  $\frac{k+3}{k+2} = 1 + \frac{1}{k+2}$ . From the asymptotic equivalence

$$\log(1 + \varepsilon(x)) \sim \varepsilon(x) \quad \text{as} \quad \varepsilon(x) \rightarrow 0$$

we have

$$\log \left( 1 + \frac{1}{k+2} \right) \sim \frac{1}{k+2} \sim \frac{1}{k} \quad \text{as} \quad k \rightarrow +\infty$$

Thus it is a series with positive terms whose general term is asymptotically equivalent to  $1/k$ . By comparison with the harmonic series, this series diverges to  $+\infty$ .

## Example

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \left( \frac{1}{k} - \sin \frac{1}{k} \right)$$

From the third-order Maclaurin expansion

$$\sin(\varepsilon(x)) = \varepsilon(x) - \frac{(\varepsilon(x))^3}{3!} + o((\varepsilon(x))^3) \quad \text{as } \varepsilon(x) \rightarrow 0$$

we have

$$\frac{1}{k} - \sin \frac{1}{k} = \frac{1}{k} - \left( \frac{1}{k} - \frac{1}{6k^3} + o\left(\frac{1}{k^3}\right) \right) \sim \frac{1}{6k^3} \quad \text{as } k \rightarrow +\infty$$

Therefore the series, which has positive terms, converges by limit comparison with the series of  $1/(6k^3)$ .

# Summary

## Passing from the discrete to the continuous:

- ▶ Asymptotic equivalences of functions also hold for sequences, taking as  $\varepsilon(x)$  an infinitesimal sequence (for example  $1/k$ )
- ▶ Notable equivalences and Maclaurin expansions provide the asymptotic term

## Strategy:

- ▶ Find a simple  $b_k$  with  $a_k \sim b_k$ , typically  $b_k = c/k^\alpha$
- ▶ Conclude with the limit comparison test and the generalized harmonic series

# Root test

# Introduction: Root test

- ▶ The **root test** compares the series with a **geometric series** by studying the limit of  $\sqrt[k]{a_k}$
- ▶ It is particularly useful when the general term contains  **$k$ -th powers**
- ▶ We will apply it to the series  $\sum k^\beta b^k$  and  $\sum b^k/k^k$

## Theorem 5: Root test

Let  $\{a_k\}$  be a sequence with non-negative terms. If the following limit exists:

$$\lim_{k \rightarrow +\infty} \sqrt[k]{a_k} = \ell \in \mathbb{R}^*$$

then the series

$$\sum a_k \quad \text{is} \quad \begin{cases} \text{convergent} & \text{if } \ell < 1 \text{ or } \ell = -\infty \\ \text{divergent to } +\infty & \text{if } \ell > 1 \text{ or } \ell = +\infty \end{cases}$$

## Proof (1/4)

Suppose that  $\lim_{k \rightarrow +\infty} \sqrt[k]{a_k} = \ell < 1$ .

Since  $\sqrt[k]{a_k} \rightarrow \ell \in \mathbb{R}$ , for every  $\varepsilon > 0$  there exists  $m \in \mathbb{N}$  such that  $\forall k \geq m$ :

$$\sqrt[k]{a_k} \leq \ell + \frac{\varepsilon}{2}$$

Moreover, since  $\ell < 1$ , we have  $\ell < 1 - \varepsilon$  for a suitable  $\varepsilon > 0$ . For this  $\varepsilon$  we therefore have, eventually:

$$\sqrt[k]{a_k} \leq \ell + \frac{\varepsilon}{2} < (1 - \varepsilon) + \frac{\varepsilon}{2} = 1 - \frac{\varepsilon}{2}$$

hence

$$a_k < \left(1 - \frac{\varepsilon}{2}\right)^k, \quad \forall k \geq m$$

## Proof (2/4)

We have thus proved that  $a_k < \left(1 - \frac{\varepsilon}{2}\right)^k$  eventually.

The geometric series:

$$\sum_{k=0}^{\infty} \left(1 - \frac{\varepsilon}{2}\right)^k \quad \text{is convergent since } 1 - \frac{\varepsilon}{2} < 1$$

Hence, by the comparison test, the original series converges.

## Proof (3/4)

Suppose that  $\lim_{k \rightarrow +\infty} \sqrt[k]{a_k} = \ell > 1$ .

Since  $\sqrt[k]{a_k} \rightarrow \ell \in \mathbb{R}$ , for every  $\varepsilon > 0$  there exists  $m \in \mathbb{N}$  such that  $\forall k \geq m$ :

$$\sqrt[k]{a_k} \geq \ell - \frac{\varepsilon}{2}$$

Moreover, since  $\ell > 1$ , we have  $\ell > 1 + \varepsilon$  for a suitable  $\varepsilon > 0$ . For this  $\varepsilon$  we therefore have, eventually:

$$\sqrt[k]{a_k} \geq \ell - \frac{\varepsilon}{2} > (1 + \varepsilon) - \frac{\varepsilon}{2} = 1 + \frac{\varepsilon}{2}$$

hence

$$a_k > \left(1 + \frac{\varepsilon}{2}\right)^k, \quad \forall k \geq m$$

## Proof (4/4)

We have thus proved that  $a_k > \left(1 + \frac{\varepsilon}{2}\right)^k$  eventually.

The geometric series:

$$\sum_{k=0}^{\infty} \left(1 + \frac{\varepsilon}{2}\right)^k \quad \text{is divergent since } 1 + \frac{\varepsilon}{2} > 1$$

Hence, by the comparison test, the original series diverges. □

The cases  $\ell = -\infty$  and  $\ell = +\infty$

Clearly, these arguments remain valid also if  $\ell = -\infty$  or  $+\infty$ , respectively.

The series  $\sum k^\beta b^k$

### Observation 5

The series  $\sum_{k=1}^{\infty} k^\beta \cdot b^k$  with  $\beta \in \mathbb{R}$ ,  $b \geq 0$  is

$$\left\{ \begin{array}{ll} \text{convergent} & \text{if } b < 1 \\ \text{divergent} & \text{if } b > 1 \\ \text{convergent} & \text{if } b = 1 \text{ and } \beta < -1 \\ \text{divergent} & \text{if } b = 1 \text{ and } \beta \geq -1 \end{array} \right.$$

## Proof

It is a series with non-negative terms, and we have:

$$\sqrt[k]{k^\beta \cdot b^k} = b \cdot k^{\beta/k}$$

and

$$\lim_{k \rightarrow +\infty} k^{\beta/k} = \lim_{k \rightarrow +\infty} \exp \left( \underbrace{\frac{\beta}{k} \cdot \log k}_{\rightarrow 0} \right) = 1$$

hence, by the root test, it converges for  $b < 1$  and diverges for  $b > 1$ . If  $b = 1$ , the series becomes:

$$\sum_{k=1}^{\infty} k^\beta = \sum_{k=1}^{\infty} \frac{1}{k^{-\beta}}$$

that is, a generalized harmonic series; it is convergent for  $\beta < -1$ , while it is divergent for  $\beta \geq -1$ . □

The series  $\sum b^k/k^k$

### Observation 6

*The series  $\sum_{k=1}^{\infty} b^k/k^k$  with  $b \geq 0$  is convergent.*

**Proof.** It is a series with non-negative terms, and we have:

$$\sqrt[k]{\frac{b^k}{k^k}} = \frac{b}{k} \rightarrow 0 \quad \text{as} \quad k \rightarrow +\infty$$

hence the series converges by the root test. □

# Summary

## Root test:

- ▶ If  $a_k \geq 0$  and  $\sqrt[k]{a_k} \rightarrow \ell$ : the series converges if  $\ell < 1$  and diverges to  $+\infty$  if  $\ell > 1$
- ▶ The proof compares  $a_k$  with a geometric series with ratio  $1 \mp \frac{\varepsilon}{2}$

## Applications:

- ▶  $\sum k^\beta b^k$ : converges if  $b < 1$ , diverges if  $b > 1$ ; if  $b = 1$  it is a generalized harmonic series
- ▶  $\sum b^k / k^k$  converges for every  $b \geq 0$

# Ratio test

# Introduction: Ratio test

- ▶ The **ratio test** studies the limit of the ratio of two consecutive terms  $a_{k+1}/a_k$
- ▶ In this case too, the series is compared with a **geometric series**
- ▶ It is particularly useful when the general term contains **factorials**: we will use it for the series  $\sum \frac{1}{k!}$

## Theorem 6: Ratio test

*Let  $\{a_k\}$  be a sequence with positive terms. If the following limit exists:*

$$\lim_{k \rightarrow +\infty} \frac{a_{k+1}}{a_k} = \ell \in \mathbb{R}^*$$

*then the series*

$$\sum a_k \quad \text{is} \quad \begin{cases} \text{convergent} & \text{if } \ell < 1 \text{ or } \ell = -\infty \\ \text{divergent to } +\infty & \text{if } \ell > 1 \text{ or } \ell = +\infty \end{cases}$$

## Proof (1/4)

Suppose that  $\lim_{k \rightarrow +\infty} a_{k+1}/a_k = \ell < 1$ . Reasoning as in the proof of the root test, there exists  $m \in \mathbb{N}$  such that  $\forall k \geq m$ :

$$\frac{a_{k+1}}{a_k} < \left(1 - \frac{\varepsilon}{2}\right)$$

for a suitable  $\varepsilon > 0$ . Reasoning iteratively, this implies that:

$$\begin{aligned} a_{k+1} &< \left(1 - \frac{\varepsilon}{2}\right) a_k < \left(1 - \frac{\varepsilon}{2}\right) \left(1 - \frac{\varepsilon}{2}\right) a_{k-1} \\ &< \dots < \left(1 - \frac{\varepsilon}{2}\right)^{k-m+1} a_m \end{aligned}$$

## Proof (2/4)

We have thus proved that  $a_k < \left(1 - \frac{\varepsilon}{2}\right)^{k-m} a_m$  eventually. The geometric series:

$$\sum_{k=m}^{\infty} \left(1 - \frac{\varepsilon}{2}\right)^{k-m} a_m \text{ is convergent since } 1 - \frac{\varepsilon}{2} < 1$$

Hence, by the comparison test, the original series converges.

## Proof (3/4)

Suppose that  $\lim_{k \rightarrow +\infty} a_{k+1}/a_k = \ell > 1$ . Reasoning as in the proof of the root test, there exists  $m \in \mathbb{N}$  such that  $\forall k \geq m$ :

$$\frac{a_{k+1}}{a_k} > \left(1 + \frac{\varepsilon}{2}\right)$$

for a suitable  $\varepsilon > 0$ . Reasoning iteratively, this implies that:

$$\begin{aligned} a_{k+1} &> \left(1 + \frac{\varepsilon}{2}\right) a_k > \left(1 + \frac{\varepsilon}{2}\right) \left(1 + \frac{\varepsilon}{2}\right) a_{k-1} \\ &> \dots > \left(1 + \frac{\varepsilon}{2}\right)^{k-m+1} a_m \end{aligned}$$

## Proof (4/4)

We have thus proved that  $a_k > \left(1 + \frac{\varepsilon}{2}\right)^{k-m} a_m$  eventually. The geometric series:

$$\sum_{k=m}^{\infty} \left(1 + \frac{\varepsilon}{2}\right)^{k-m} a_m \text{ is divergent since } 1 + \frac{\varepsilon}{2} > 1$$

Hence, by the comparison test, the original series diverges. □

The cases  $\ell = -\infty$  and  $\ell = +\infty$

Clearly, these arguments remain valid also if  $\ell = -\infty$  or  $+\infty$ , respectively.

The series  $\sum 1/k!$

### Observation 7

$$\sum_{k=0}^{\infty} \frac{1}{k!} = e$$

**Proof.** We have

$$\frac{1/(k+1)!}{1/k!} = \frac{1}{k+1} \rightarrow 0 \quad \text{as } k \rightarrow +\infty$$

Therefore the series, which has non-negative terms, converges by the ratio test. The proof that the sum of the series is equal to  $e$  will be given later using series of functions. □

# Summary

## Ratio test:

- ▶ If  $a_k > 0$  and  $a_{k+1}/a_k \rightarrow \ell$ : the series converges if  $\ell < 1$  and diverges to  $+\infty$  if  $\ell > 1$
- ▶ Iterating the estimate on the ratio,  $a_k$  is compared with a geometric series

## Application:

- ▶ The series  $\sum_{k=0}^{\infty} \frac{1}{k!}$  converges (and its sum is  $e$ )

# Series with non-positive terms

# Introduction: Series with non-positive terms

- ▶ The tests seen so far require non-negative terms
- ▶ We see that they easily extend to series whose terms **eventually** have the same sign

## Extension of the tests

- ▶ We know that the behavior of a series does not change if we alter a finite number of its terms. Consequently, the tests for series with non-negative terms also apply to series whose terms are eventually non-negative.
- ▶ Moreover, by factoring a minus sign out of the whole series, we see that these tests can also be applied to series with non-positive terms, and hence to series whose terms are eventually non-positive.

In summary, therefore, the tests we have seen apply to series all of whose terms (except for a finite number) have the same sign.

## Series with terms of constant sign:

- ▶ Altering a finite number of terms does not change the behavior of a series
- ▶ By factoring out a minus sign, the tests also hold for series with non-positive terms
- ▶ The tests apply to series all of whose terms, except for a finite number, have the same sign

# Key messages

# Summary (1/2)

## 1. Series with non-negative terms:

- ▶ They are regular: they converge if and only if the partial sums are bounded

## 2. Comparison tests:

- ▶ Comparison:  $0 \leq a_k \leq b_k$  eventually, convergence passes from the dominating series and divergence from the dominated series
- ▶ Limit comparison:  $a_k \sim b_k \Rightarrow$  same behavior (not the same sum)

## 3. Condensation test:

- ▶ For decreasing  $\{a_k\}$ :  $\sum a_k$  and  $\sum 2^k a_{2^k}$  have the same behavior

## Summary (2/2)

### 4. Generalized harmonic series:

- ▶  $\sum \frac{1}{k^\alpha}$  diverges if  $\alpha \leq 1$  and converges if  $\alpha > 1$

### 5. Root and ratio tests:

- ▶ Limit  $\ell$  of  $\sqrt[k]{a_k}$  or of  $a_{k+1}/a_k$ : convergence if  $\ell < 1$ , divergence if  $\ell > 1$

### 6. Extension:

- ▶ All the tests hold for series whose terms eventually have the same sign